

PROJECT & RESEARCH INQUIRY GUIDE

Start with the problem, evidence, and decision that needs support.

BAITSSS can be part of water-management, research, consulting, agency, and university work. A first inquiry does not need to decide whether the next step is software access, project analysis, training, research collaboration, or integration. The useful starting point is the technical problem.

Send four things.

01 PLACE

Field, farm, district, basin, program area, research site, or another study area.

02 PERIOD

Single event, growing season, year, multi-year study, drought period, or another time window.

03 QUESTION

What needs to be understood, compared, estimated, investigated, verified, or supported?

04 EVIDENCE

Measurements, irrigation or pumping records, weather, field observations, satellite products, and agency records.

Examples of questions that can be discussed

- How did a field respond to reduced water supply or a change in irrigation timing?
- Can measured deliveries, pumping, satellite estimates, and modeled field behavior be interpreted together?
- What do soil-water and irrigation dynamics show that a seasonal ET total does not?
- How might agricultural demand or field response change under conservation, drought, recycled-water, or allocation scenarios?

WHERE BAITSSS CAN ADD VALUE

Use the existing water system. Add the field-process layer where useful.

BAITSSS does not require an organization to replace its meters, weather network, GIS, remote-sensing products, agency records, groundwater model, delivery system, or existing reporting workflow. Those systems can remain authoritative for the roles they already serve.

BAITSSS can contribute

- Hourly field-process simulation
- Evapotranspiration and component analysis
- Surface and root-zone soil-water states
- Irrigation behavior and seasonal water demand
- 30 m spatial variability and mapped Results
- Management and conservation scenarios
- Method comparison and independent evaluation
- Reproducible technical records and exports

Potential project settings

- Conservation and allocation review
- Irrigation-management analysis
- Agricultural-demand questions
- Groundwater-demand inputs
- Recycled-water and drought scenarios
- Verification and difficult-field diagnosis
- Research experiments and method comparison
- Consulting, agency, and university studies

What happens after the first inquiry

1. Understand the question

Clarify the place, period, decision, available evidence, and systems already in use.

2. Define the analytical role

Identify what BAITSSS can add and what should remain with existing measurements, models, or operational systems.

3. Scope the work

The next step may be guidance, a demonstration, project analysis, research collaboration, training, software use, or a broader organizational discussion.

For substantial student support, validation, benchmarking, or source-code questions, including the advisor or project lead helps clarify the research objective and an appropriate arrangement.